Tuan LE

Machine Learning & Perception Engineer · Predictive Maintenance, Anomaly Detection, Computer Vision

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PROFILE

Machine-learning researcher (PhD, Université de Technologie de Compiègne, 2024) who builds models that learn from few labels and run on real hardware. During my PhD at the Roberval Lab I worked on anomaly detection, control-chart monitoring, quality forecasting and explainable defect detection for manufacturing sensors and images. As a postdoc at Heudiasyc (CNRS), I build LiDAR–camera 3D perception for autonomous vehicles and deploy perception models on a ROS 2 vehicle and an FPGA edge board. I am looking for ML or perception engineer roles in predictive maintenance, inspection or autonomous systems.

CORE EXPERTISE

Condition monitoring:	Unsupervised anomaly and fault detection on multivariate sensor time series (vibration, temperature); noise-robust autoencoders; benchmarks on NASA SMAP/MSL and SMD
PHM and RUL:	Remaining-useful-life prediction and degradation modelling on NASA C-MAPSS, with seed-level and engine-level uncertainty reporting
Process and quality:	SPC control-chart pattern prediction (CNN); quality forecasting (peephole LSTM); defect detection on thermal images with explainable AI (XAI)
3D perception:	3D object detection from LiDAR and cameras (CenterPoint, TransFusion), LiDAR–camera fusion, open-vocabulary detection with vision-language models (GroundingDINO, SAM, CLIP), event cameras
Embedded and edge:	INT8 on FPGA (Vitis AI DPU), binary neural networks, Prophesee event camera, real-time ROS 2 integration, FP16 inference, latency profiling

RECENT PROJECT

Remaining-useful-life prediction for turbofan engines (NASA C-MAPSS) 2026 · github.com/nautel/cmapss-rul-miras

- Compared literature baselines (DCNN, re-implemented STA-HPINN) with test-time-memory sequence models from Google's Miras framework (Titans, Memora, Mamba2, DeltaNet) in one shared pipeline: about 2,900 training runs, 4 subsets, 10 seeds per result.
- Scored with engine-level bootstrap 95% CIs and paired tests. Found that CIs (about ±2 RMSE) are wider than most gaps between methods and that two published results could not be reproduced.
- Ablations: operating-condition normalisation mattered 3.4× more than architecture (removing it cost 1.37 RMSE versus 0.40); dropping the physics-informed loss of STA-HPINN improved RMSE by 1.32.

EXPERIENCE

Postdoctoral Researcher, Perception for Autonomous Systems 03/2025 – present

Heudiasyc Lab (CNRS UMR 7253), Université de Technologie de Compiègne · Advisors: J. Moreau, F. Davoine

- **Open-vocabulary 3D detection.** Generated 3D labels for new object classes by grounding vision-language models (GroundingDINO, SAM, CLIP) in LiDAR point clouds, with no manual 3D annotation; an analytical fitting formulation made label generation more than 10× faster than iterative search (video: nautel.github.io). Trained and benchmarked CenterPoint and TransFusion detectors on nuScenes, KITTI and Lyft on a multi-GPU SLURM cluster.
- **Real-time and edge deployment.** Deployed the detector as a ROS 2 node on the lab vehicle (FP16, C++ pre-processing) and validated it end to end with latency measured for each stage. Ran a live INT8 demo on a Xilinx Kria KV260 FPGA (Vitis AI DPU) with a Prophesee GenX320 event camera; raised throughput from 8 to 27 FPS by profiling each stage.
- **Efficient models (REVE-BNN project, since 07/2026).** Binary neural networks for event-camera detection (Gen1, 1 Mpx) on robots with a tight power budget; measured how detection degrades when the inference rate drops from 20 Hz to 5 Hz, and reproduced published baselines from scratch.

PhD Researcher, Anomaly Detection & Process Monitoring for Smart Manufacturing 2021 – 2024

Roberval Lab, Université de Technologie de Compiègne · Advisors: H. C. Vu, Z. Cherfi-Boulanger

- **Machine-signal fault detection (RCLED).** Designed an autoencoder combining convolution, LSTM and self-attention layers, with a training scheme derived from robust PCA that separates noise from normal patterns, so no anomaly labels are needed. On vibration and temperature data from an industrial machine (Sensoteq KappaX condition-monitoring sensor) it reached an F1 score of 0.85 (*Neurocomputing*, 2024; code: github.com/nautel/RCLED).
- **Control-chart pattern prediction.** Adapted a CNN to predict control-chart patterns in simulated SPC time series, covering six process states (normal, cyclic, upward and downward trends and shifts), so that abnormal states can be anticipated before defects occur; tuned kernel size, filters and dense layers against MAE (ESREL 2022).
- **Product-quality forecasting.** Forecast non-linear, non-stationary quality series over long horizons with peephole LSTMs, outperforming the compared forecasting methods (CIGI QUALITA MOSIM 2023).
- **Explainable inspection of fuel tanks.** Detected and localised non-conforming fuel tanks in thermal images, with an explainable-AI approach to show what drives each detection (MIMAR 2023).
- **Synthetic anomalies.** Designed a latent-space augmentation that synthesises logical anomalies by perturbing image features, improving detection at no extra inference cost.

Electrical Engineer 2019 – 2021

VATEC Energy Consulting (2021) · Vietnam Electricity EVN, internship (2020) · Da Nhim hydropower plant, 240 MW, internship (2019)

- Wind-resource geospatial pipeline in Python for wind-farm site selection (VATEC); PyPSA grid-expansion model for Vietnam's 500 kV grid (EVN); generator operation monitoring (Da Nhim).

PUBLICATIONS

1. **T. Le**, H. C. Vu, A. Ponchet-Durupt, N. Boudaoud, Z. Cherfi-Boulanger, T. Nguyen-Trang. Unsupervised detecting anomalies in multivariate time series by Robust Convolutional LSTM Encoder–Decoder (RCLED). **Neurocomputing** 592, 127791, 2024. doi:10.1016/j.neucom.2024.127791
2. **T. Le**, H. C. Vu, A. Ponchet-Durupt, N. Boudaoud, Z. Cherfi-Boulanger. Forecasting product quality using peephole long short-term memory. **CIGI QUALITA MOSIM 2023**, Trois-Rivières, Canada. doi:10.60662/8gn5-5575 Oral
3. **T. Le**, H. C. Vu, N. Boudaoud, Z. Cherfi-Boulanger, A. Ponchet-Durupt, H. S. H. Nguyen. A deep learning approach for control chart patterns (CCPs) prediction. **ESREL 2022**, European Safety and Reliability Conference, Dublin, Prognostics and System Health Management session. doi:10.3850/978-981-18-5183-4_R22-09-265 Oral
4. **T. Le**, J. Moreau, F. Davoine. Open-vocabulary 3D object detection via multi-stage analytical fitting. Under review, 2026.
5. M. Q. Duong, H. H. Nguyen, **T. Le**, M. Mussetta. New planning for the 500 kV Vietnamese grid with high penetration of renewable energy sources. **IEEE Milan PowerTech**, 2019. doi:10.1109/PTC.2019.8810879

TALKS

- **Explainable Artificial Intelligence (XAI) for non-conforming product detection: an application to fuel tank manufacturing.** 12th IMA International Conference on Modelling in Industrial Maintenance and Reliability (MIMAR), University of Nottingham, UK, 2023.
- **Advanced predictive maintenance models based on machine learning for Industry 4.0.** Roberval Lab doctoral days, UTC: oral (2023) and poster (2022).

EDUCATION

PhD in Applied Machine Learning, Université de Technologie de Compiègne, France. Thesis: *Semi-supervised Deep Learning for Robust Anomaly Detection in Smart Manufacturing Systems*. 2021 – 2024

Engineering Degree in Electrical Engineering, Danang University of Technology, Vietnam. Power systems specialisation, GPA 3.4 / 4.0. 2016 – 2021

SKILLS & LANGUAGES

Machine learning:	PyTorch (daily), LSTM / ConvLSTM, autoencoders, CNNs, transformers and state-space models, semi-supervised and self-supervised learning, scikit-learn
Perception and edge:	LiDAR–camera fusion, point-cloud processing (Open3D), segmentation, vision-language models (GroundingDINO, SAM, CLIP), event cameras, INT8 quantisation (Vitis AI), CUDA, torch.compile; TensorRT, ONNX Runtime (working knowledge)
Engineering:	Python, C/C++, Matlab, ROS 2, MMDetection3D, PyPSA, SLURM multi-GPU, Docker, Git, Linux
Languages:	Vietnamese (native), English (fluent), French (B1, daily working language in the lab)